Split Among Selected Members

Percentage Split

Receipt Upload

FAIRSHARE

Transparent Expenses, Hassle-Free Settlements

SPRINT 2 PRESENTATION

KHUSHI SHAH

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SPRINT 2 - SMART SPLIT OPTIONS

Enhancing the expense splitting experience with flexible participant selection and custom percentage-based calculations.

Select Participants

Choose specific group members for each expense rather than forcing all participants to contribute.

Custom Percentages

Custom Percentages
Split expenses by exact percentages reflecting each person's actual share of the cost.

Attach Receipts

Upload images or PDFs to provide proof and reference for every transaction.

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USER STORY 1: SELECT PARTICIPANTS FOR AN EXPENSE

BENEFIT

So that only the people involved in the expense are required to pay, instead of forcing all group members to participate

FEATURE

I want to choose which members share a particular expense

ROLE

As a user managing shared expenses in a group

VALUE

Customized Fairness: Allows for flexible group dynamics where not every member is responsible for every transaction, upholding the "Fair Share" principle.

STORY POINTS

5

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USER STORY 2: PERCENTAGE-BASED EXPENSE SPLIT

BENEFIT

So that each participant pays an amount proportional to their actual share of the expense.

FEATURE

I want to split an expense by custom percentages

ROLE

As a user managing shared costs

VALUE

Proportional Fairness: Ensures that members with different consumption levels or budgets only pay their fair share based on an agreed-upon ratio, such as a 20%/50%/30% split.

STORY POINTS

8

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USER STORY 3: ATTACH RECEIPTS TO EXPENSES

BENEFIT

So that group members have proof and reference of the transaction.

FEATURE

I want to attach a receipt (image or PDF) to an expense

ROLE

As a user recording expenses

VALUE

Verification & Trust: Allows group members to personally audit the transaction details, such as items purchased and tax applied, fostering transparency.

STORY POINTS

8

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SPRINT 2 - KEY METRICS

- Planned Velocity: 21 points
- Actual Velocity - 21 points
- Sprint Predictability Rate: 100%
- Sprint Addition Rate: 0%
- Sprint Completion Rate: 100%
- Actions have been evaluated and kept

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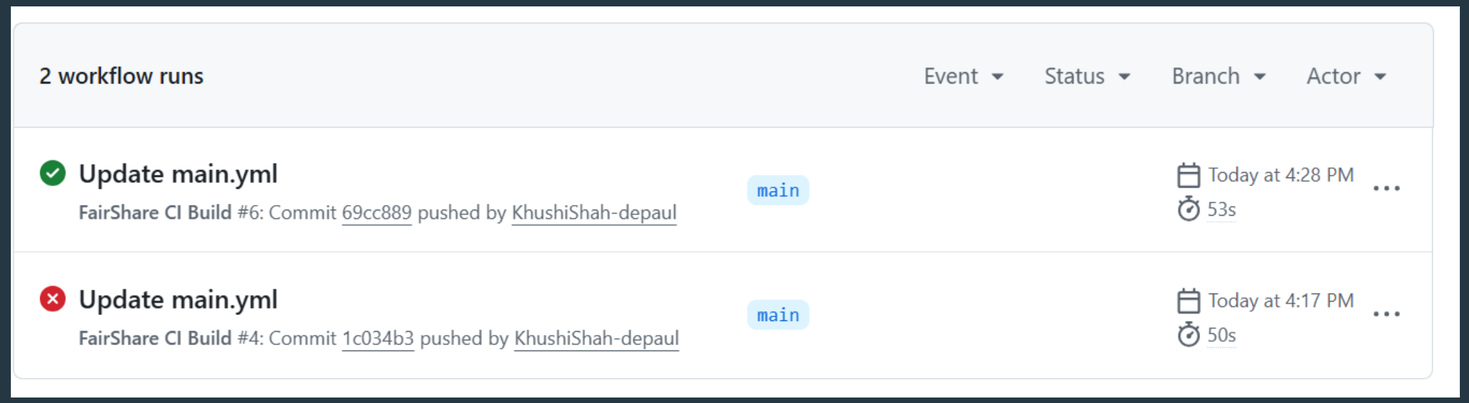
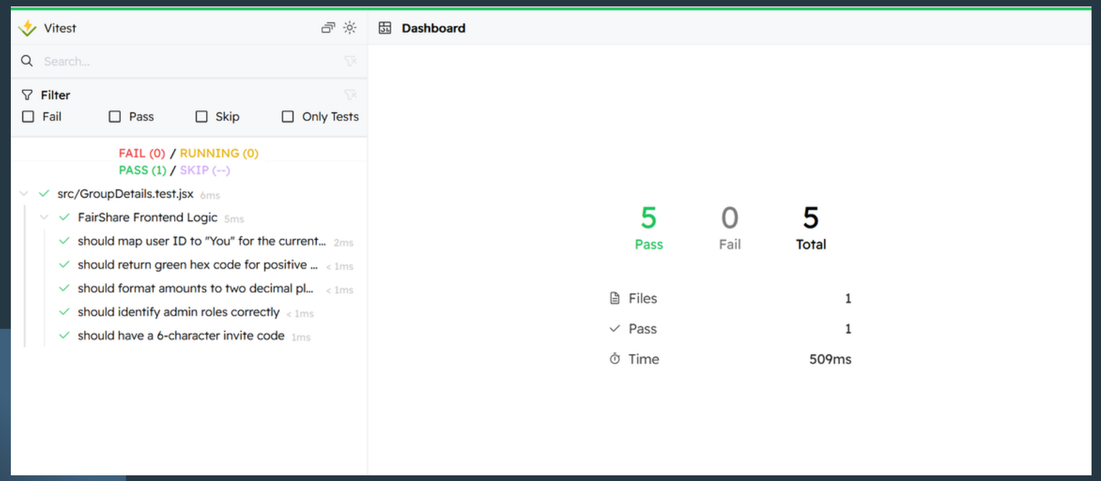
TESTS & GITHUB ACTIONS

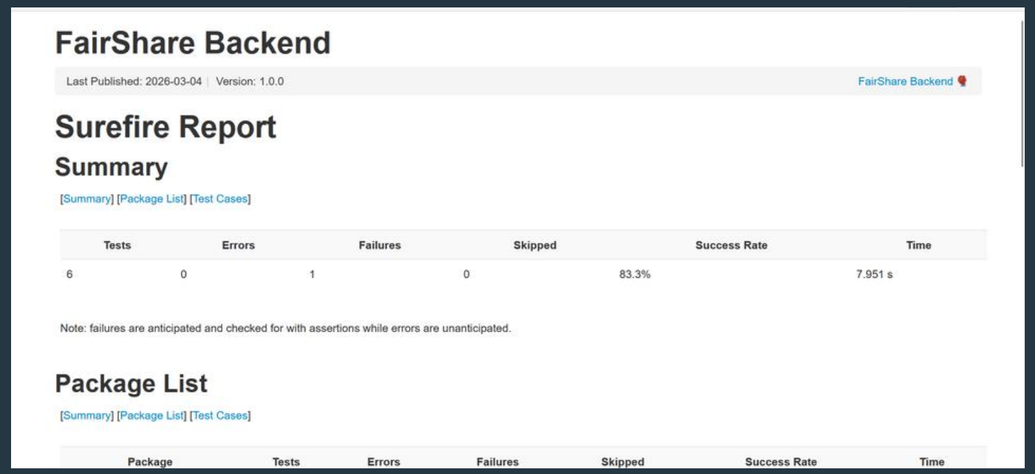


CI/CD Pipeline (GitHub Actions)

- Unit Test Automation: Every push triggers a full suite of Maven (Backend) and Vitest (Frontend) tests to ensure logic integrity.
- Build Resilience: Highlighting our troubleshooting process—moving from initial Failed Builds (dependency/logic errors) to a consistent Successful Green Tick state.



Frontend Tests



Backend Tests

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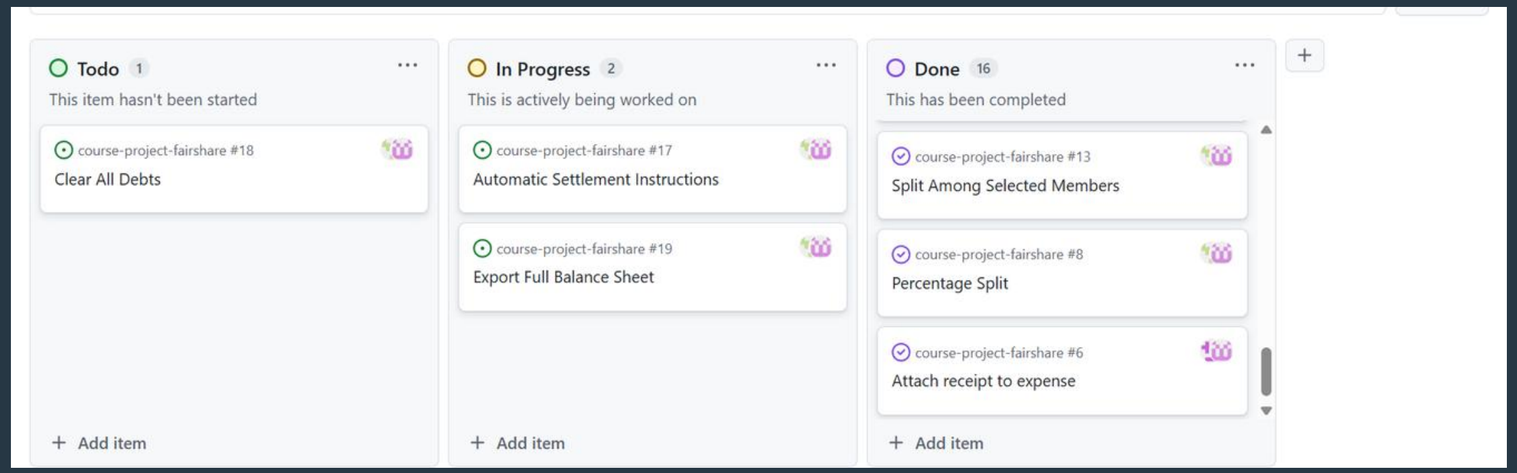
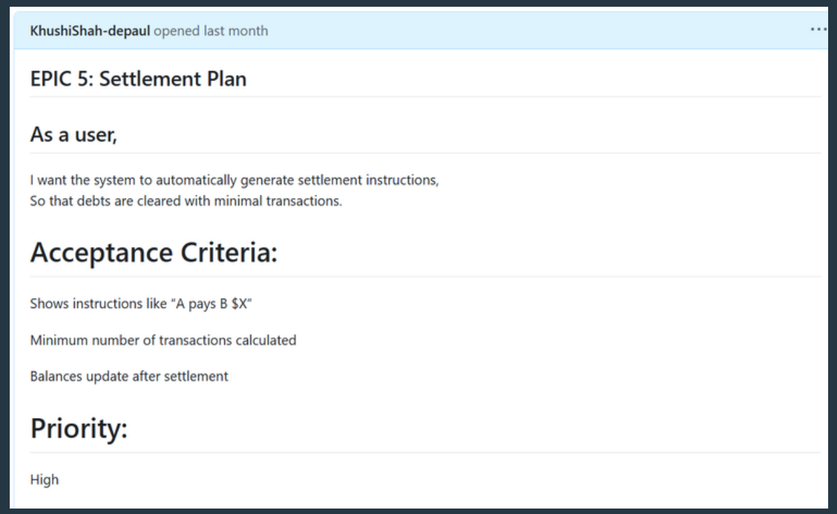
RELEASE PLANNING BOARD



Planning Board

- Sprint 1 and Sprint 2 tasks have been completed
- Sprint 3 : Automatic Settlement Instructions and Export Full Balance Sheet is in progress
- Sprint 3: Clear all debts, in still pending to be done

Issues have been created on github with user story and acceptance criteria for sprint 3



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SPRINT 3 - SETTLEMENT PLAN

01. Automated Settlement Instructions

System generates optimal payment instructions to clear all debts with minimal transactions.

02. Clear All Debts

Mark settlements as complete and reset group balances to zero after payments are made.

03. Export Balance Sheet

Download complete transaction and balance records as CSV or PDF files.

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SPRINT 3-STORIES (READY TO WORK & TASK BREAKDOWN)

Automated Settlement Instructions

User Story:
As a user, I want the system to automatically generate settlement instructions so that debts are cleared with minimal transactions.

Tasks
- Design settlement calculation logic
- Implement debt simplification algorithm
- Generate instructions like “A pays B $X”
- Display settlement plan in UI

Story Points: 8
(High complexity – algorithm + UI logic)

Clear All Debts / Reset Balances

User Story:
As a user, I want to clear all debts after payment so that my balance sheet reflects zero owed.

Tasks
- Implement “Mark as Settled” functionality
- Reset all group balances to zero
- Update dashboard balance state
- Maintain settlement history record

Story Points: 5
(Moderate complexity – backend logic + dashboard update)

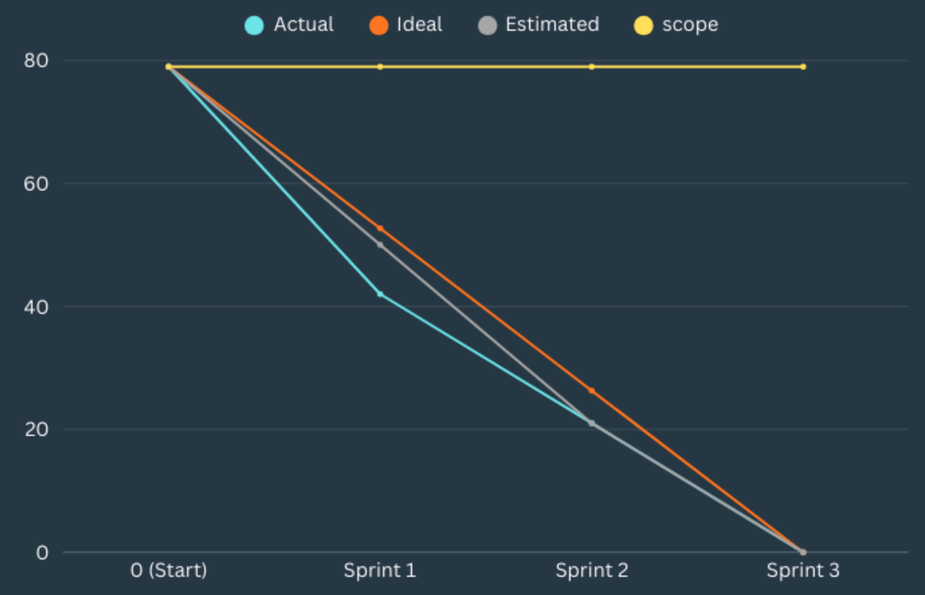
Export Balance Sheet

User Story:
As a user, I want to export the full balance sheet so that I can keep a record of all transactions and balances.

Tasks
- Generate balance sheet data from backend
- Implement CSV/PDF export functionality
- Ensure exported data matches dashboard calculations
- Provide download option in UI

Story Points: 3
(Lower complexity – export feature)

BURNDOWN CHART



- Scope Stability (Yellow Line): Total project scope remained fixed at 79 story points, demonstrating disciplined requirement management and zero scope creep.
- Performance Velocity (Blue Line): The team is currently ahead of schedule, with 58 points completed and only 21 points remaining at the end of Sprint 2.
- Efficiency vs. Baseline (Orange Line): Actual progress has consistently tracked below the Ideal Burndown, proving that the initial architectural investments are paying off in delivery speed.
- Projected Completion (Grey Line): Based on our current median velocity, the Estimated Forecast confirms a 100% feature-complete status by the end of Sprint 3.

PROJECT VELOCITY & ESTIMATION

METRIC	VALUE
Total Project Scope	79 Story Points
Completed (Sprint 1 + 2)	58 Story Points
Remaining Backlog	21 Story Points (Sprint 3)
Current Median Velocity	29 Points / Sprint

We have already completed 79% of the work (58 points) across two sprints, our velocity is actually quite high.

The 90% Confidence Interval (CI) Forecast

- We finish Sprint 3 ahead of schedule, allowing for extra "polishing" or stretch goals.
- We are on track to be 100% Feature Complete by the end of Sprint 3.

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RETROSPECTIVE

START

- Modular Math Services: Start extracting the splitting logic (Equal and Percentage) into dedicated, reusable service classes to keep the controllers clean and maintainable.
- Edge-Case Unit Testing: Start writing specific test cases for 0.00 balances and rounding errors (e.g., $1/3$ splits) that weren't fully covered in the initial unit test suite.
- Peer Code Reviews: Implementing mandatory second-eye reviews on GitHub to catch logic errors in complex split calculations before they reach the CI pipeline.

CONTINUE

- GitHub Actions CI Integration: Continue the practice of blocking merges until the Maven and Vitest suites return a 100% green pass rate.
- Receipt Validation Logic: Continue using the magnifying-glass audit trail logic for the "Valid" stamp, as it has proven to be a robust way to verify file uploads.
- Automated Testing Rigor: Maintaining the 100% pass rate for Maven and Vitest unit tests.
- High-Velocity Collaboration: Continuing the transparent communication that allowed the team to burn down 21 points while maintaining scope stability.

STOP

- Hardcoding Test Data: Stop using static user IDs in backend tests; transition to dynamic mock data to avoid the environment-specific failures seen in the earlier build attempts.
- Commiting Without Local Linting: Stop pushing code that hasn't been run against the local linter to reduce "noise" and failed builds in the GitHub Actions pipeline.
- Last-Minute Environment Fixes: Stopping the practice of debugging CI/CD configuration during the final hours of the sprint.

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THANK YOU

FOR YOUR NICE ATTENTION

Demonstration Video: https://youtu.be/c3UXRfEIc08